

Crypto Ransomware Detection

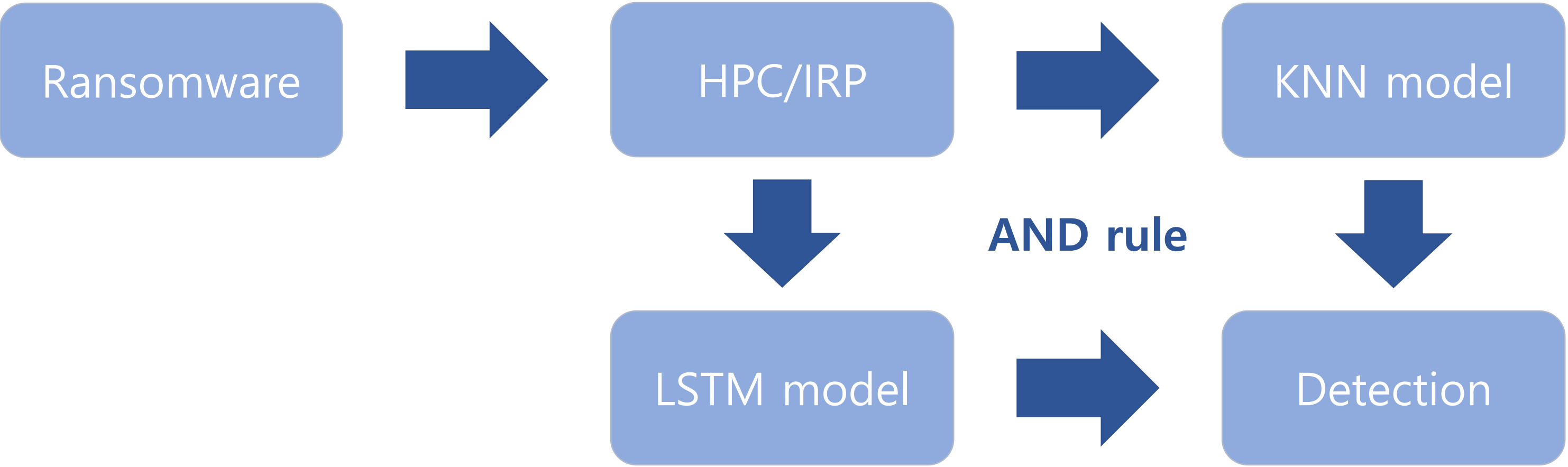
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Introduction

Topic	Ransomware
Definition	Malware, Encryption, Demand
Goal	Detect Ransomware using ML

Theoretical Background

Base study: RansomRadar



HPC counter	Tracks low-level hardware activities, CPU execution Measured using Windows Performance Toolkit
IRP log	Tracks kernel-mode data structures, I/O behaviors Measured using WDK and minifilter driver

Methods

1. Reproduction setup

Windows 11	Intel i7 CPU	Hyper-V VM	ADK/WDK/SDK
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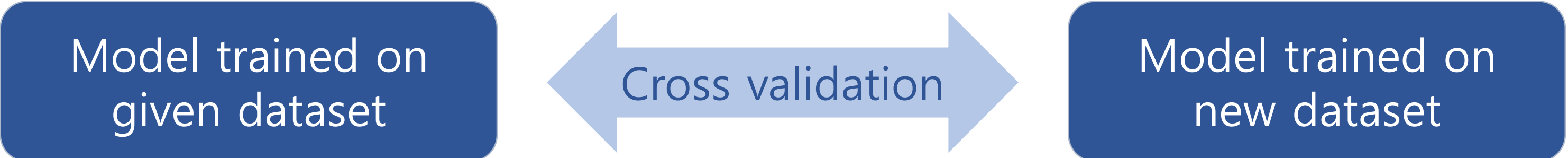
2. Dataset generation

Benign	10 min	MS Edge, Bandizip, git/pip/cargo, ffmpeg, 7zip, browser streaming, Vera Crypt, GPG, wget, Idle
Ransomware	2 min	First try 2 min for encryption
	10 min	Try again with 10 min duration
Encryption check	30 files of image/ppt/pdf → 5 directories (Desktop/Documents/Downloads/Users/Program Files)	

3. Benign/Ransomware detection test

Benign Data	10 feature files, 159 unique processes, 221,373 rows
Ransomware Data	33 feature files, 33 unique processes, 6,577 positive rows

4. 5-fold training & Cross-dataset validation



5. Adding new features: Write entropy



Results

Given model's evaluation				
Evaluation	Accuracy	Precision	Recall	F1
Paper's claim	97.97%	92.36%	100.0%	0.9603
Paper → Given	99.24%	96.60%	97.26%	0.9693

5-fold training & cross-dataset validation				
Evaluation	Accuracy	Precision	Recall	F1
Given → Given	98.40%	89.16%	99.04%	0.9378
New → New	0.9821	70.13%	94.55%	0.7925
Given → New	97.30%	71.18%	32.73%	0.4445
New → Given	90.04%	59.67%	69.32%	0.6302

Validation with new feature: Write entropy				
Total model	Accuracy	Precision	Recall	F1
Control	96.00%	54.40%	96.00%	0.6605
Use write entropy	96.60%	71.96%	91.00%	0.7333
LSTM only	Accuracy	Precision	Recall	F1
Control	72.53%	11.58%	100.0%	0.2055
Use write entropy	84.80%	63.21%	95.00%	0.7107

Conclusion

Findings	- Given model shows great F1 on the given dataset but drops on the new dataset. - Most of the given data were used for training the model. - Cross validation showed poor generalization of models. - Adding a new feature, write entropy, enhanced the LSTM model's F1 and performance.
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RansomRadar appears dataset-bound rather than broadly generalizable

Adding a new feature of write entropy was very successful

References

- ① Guo, Lihua, et al. "Ransomware Detection through Temporal Correlation between Encryption and I/O Behavior." Proceedings of the ACM on Software Engineering 2.FSE (2025): 197-218.
- ② van Sloun, Christian, et al. "Detecting Ransomware Despite I/O Overhead: A Practical Multi-Staged Approach." NDSS. 2025.
- ③ Hitaj, Dorjan, et al. "Minerva: A file-based ransomware detector." Proceedings of the 20th ACM Asia Conference on Computer and Communications Security. 2025.